

Building a RAG Application with LangChain

What is LangChain?

LangChain is a framework for building applications around language models. It provides components for document loading, text splitting, embeddings, vector stores, retrievers, prompts, and model interaction.

Document Loading

A RAG application first needs to load its knowledge sources. These sources may include PDF files, text files, web pages, databases, or other documents. Each loaded document can contain page content and metadata.

Text Splitting

Large documents are commonly divided into smaller chunks before embedding. Chunking helps retrieval return focused pieces of information and prevents unnecessarily large amounts of context from being sent to a language model.

Retrieval and Generation

During a query, the retriever searches the vector store and returns relevant chunks. Those chunks are inserted into a prompt along with the user's question. The language model then uses the retrieved context to generate the final response.

Practical Example

A developer could build a technical documentation assistant using LangChain, an embedding model, and Qdrant. Documentation would be chunked and indexed once. Later, user questions would retrieve the most relevant documentation before response generation.